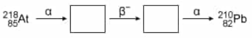
What are the products of the alpha (α) and beta (β⁻) decays, respectively, in the nuclear decay chain below?

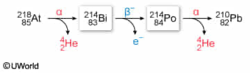


- A. $^{222}_{87}Fr$; $^{222}_{86}Rn$ [3%]
- B. $^{214}_{87}Fr$; $^{214}_{86}Rn$ [3%]
- C. $^{222}_{83}Bi$; $^{222}_{84}Po$ [5%]
- ✔ D. $^{214}_{83}Bi$; $^{214}_{84}Po$ [87%]

Correct
87% answered correctly

Explanation:

Nuclear decay chain of astatine-218



Radioactivity is the spontaneous emission of radiation (photons or particles) from an unstable atomic nucleus. Although several types of nuclear radiation exist, the three most common types are alpha decay, beta decay, and gamma emission.

During alpha (α) decay, an unstable nucleus ejects an α particle consisting of 2 protons and 2 neutrons (a helium-4 nucleus without its electrons). Accordingly, the resulting nucleus formed following an α decay will have a mass number that is **4 units less** and an atomic number that is **2 units less** than the nucleus that underwent the α decay.

Beta decay can occur in three forms: β^- decay (electron emission), β^+ decay (positron emission), and electron capture. In all three forms of beta decay, the **mass number is unchanged** but the **atomic number** either **increases (β^- decay)** or **decreases (β^+ decay)** by **1 unit**.

Consequently, the α decay of an atom of astatine-218 would form bismuth-214 by the following nuclear reaction:

$$^{218}_{85}At \rightarrow \, ^{214}_{83}Bi + \, ^{4}_{2}He$$

Subsequent β^- decay of the ^{214}Bi nucleus would eject a nuclear electron (e^-) and yield a polonium nucleus with the same mass number:

$$^{214}_{83}Bi \rightarrow \, ^{214}_{84}Po + e^-$$

Therefore, the products of the alpha (α) and beta (β^-) decays in the given nuclear decay chain of ^{218}At are ^{214}Bi and ^{214}Po, respectively.

(Choices A, B, and C) In an α decay, both the mass number and the atomic number should *decrease* by four and two units, respectively. In a β^- decay, the atomic number *increases* by one unit.

Educational objective:
In α decay, the mass number of an atomic nucleus decreases by 4 units and the atomic number decreases by 2 units. In all forms of beta decay, the mass number of an atomic nucleus remains unchanged but the atomic number increases (β^- decay) or decreases (β^+ decay) by 1 unit.

Time spent: 0 sec	**QID:** 402674
Subject:	**System:**
General Chemistry	Atomic Theory and Chemical Composition